ITA0608 - MACHINE LEARNING

LAB PROGRAMS

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Experiment 6: Naïve Bayes

Algorithm Aim

To implement the Naïve Bayes algorithm in Python and evaluate its performance using a confusion matrix and accuracy score.

Procedure

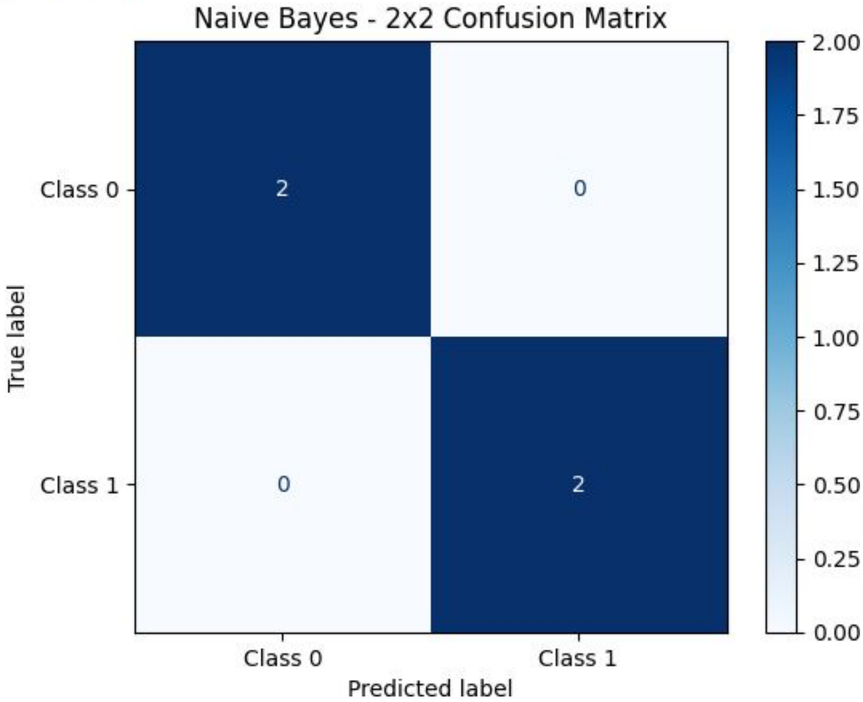
1. Import the required libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Train the Naïve Bayes classifier.
5. Predict the test data.
6. Display the confusion matrix and accuracy.

Program

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay, accuracy_score
import matplotlib.pyplot as plt

X = [
    [2,3],[3,4],[4,5],[5,6],
    [6,7],[7,8],[8,9],[9,10],
    [10,11],[11,12],[12,13],[13,14]
]

y = [0,0,0,0,0,0,1,1,1,1,1,1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42, stratify=y
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)

cm = confusion_matrix(y_test, y_pred, labels=[0,1])

print("\nConfusion Matrix:")
print(cm)

print("\nAccuracy:", accuracy_score(y_test, y_pred))

disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=["Class 0", "Class 1"])
disp.plot(cmap="Blues")

plt.title("Naive Bayes - 2x2 Confusion Matrix")
plt.show()
```

Output

```
Actual Labels: [1, 1, 0, 0]
Predicted Labels: [1 1 0 0]

Confusion Matrix:
[[2 0]
 [0 2]]

Accuracy: 1.0
```



Result

Thus, the Naïve Bayes algorithm was implemented successfully, and its performance was evaluated using the confusion matrix and accuracy score.